

STA9797 Final Project Report

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Part I - Introduction

Background Information

Statistics in sports has had a long history together that can be found in any professional sport. Metrics and game statistics have traditionally been used to evaluate player and team performances, helping determine who is better. The use of analytics in the sports industry has become immensely popular in recent years, especially in a predictive capacity. Every professional sports team today makes data-driven decisions to some capacity for a competitive advantage, baseball is no exception to this claim. In today's modern Major League Baseball (MLB) games, predictive models are used for example to generate player lineups based on the matchup for the day. Every game played generates data to be analyzed, and with the evolution of AI/ML, the era of sports analytics will only continue to grow.

Objective of Study

The primary goal of this project is to infer the key determinants of victory in Major League Baseball matchups, with a specific focus on the New York Yankees. The methodology will involve creating a predictive model using logistic regression, but the primary goal is to utilize the predictive framework as a means of explanatory modeling. The primary response variable of interest in the data is the matchup outcome for the New York Yankees, a win or loss. Given the high degree of multicollinearity inherent in baseball statistics, we employ predictive regularization techniques not only to forecast outcomes, but to stabilize coefficient estimates and perform robust variable selection. Lastly, the notion of 'home field advantage' will be explored to assess its impact on game outcomes and to determine whether playing at home significantly influences the Yankees' likelihood of winning.

Part II - Data Overview

The dataset consists of gamelogs from the 2021-2025 MLB seasons, available from [Retrosheet](#). Focusing in on the New York Yankees (NYY) matchups, the data consists of 56 features and 820 observations. The original data has been cleaned, modified and feature engineered for a more robust analysis. The original data information can be found [here](#), provided by *Retrosheet*. For our analysis, the table below shows the data after all the pre-processing.

Variable Name	Value	Type
Date	YYYY-MM-DD	date
Opponent	opposing team identifier	string
Win	1 (win), 0 (loss)	binary
home_game	1 (home), 0 (away)	binary

(team/opp)_at_bat	numerical game statistic	numerical
(team/opp)_hits	numerical game statistic	numerical
(team/opp)_score	numerical game statistic	numerical
(team/opp)_doubles	numerical game statistic	numerical
(team/opp)_sac_hits	numerical game statistic	numerical
(team/opp)_sac_flies	numerical game statistic	numerical
(team/opp)_hit_by_pitch	numerical game statistic	numerical
(team/opp)_walks	numerical game statistic	numerical
(team/opp)_intentional_walks	numerical game statistic	numerical
(team/opp)_strikeouts	numerical game statistic	numerical
(team/opp)_stolen_bases	numerical game statistic	numerical
(team/opp)_caught_stealing	numerical game statistic	numerical
(team/opp)_grounded_DP	numerical game statistic	numerical
(team/opp)_awarded_first	numerical game statistic	numerical
(team/opp)f_left_on_base	numerical game statistic	numerical
(team/opp)_pitchers_used	numerical game statistic	numerical
(team/opp)_team_earned_runs	numerical game statistic	numerical
(team/opp)_wild_pitches	numerical game statistic	numerical
(team/opp)_balks	numerical game statistic	numerical
(team/opp)_putouts	numerical game statistic	numerical
(team/opp)_assists	numerical game statistic	numerical
(team/opp)_errors	numerical game statistic	numerical
(team/opp)_passed_balls	numerical game statistic	numerical
(team/opp)_double_plays	numerical game statistic	numerical
(team/opp)_triple_plays	numerical game statistic	numerical

Each numerical game statistic appears twice as a predictor, one designated for the Yankees (**team_**) and the other designated for the opposing team (**opp_**) in the matchup. These predictors can be broken down into different functions of measurements in a game.

Category	Variable (prefix team_ / opp_)	Function
Offensive Production	at_bat, hits, score, doubles, stolen_bases	Measures ability to gain bases and score runs.
Pitching & Defensive Production	walks, intentional_walks, strikeouts, hit_by_pitch	Measures defensive efficiency and pitching stability.
Plate Discipline	sac_hits, sac_flies, caught_stealing	Measures control of the strike zone.
Play Strategy	pitchers_used, earned_runs, wild_pitches, balks, errors, passed_balls, double_plays, putouts	Measures tactical plays to advance runners.

An outlier analysis was performed on the dataset by applying Mahalanobis distance to identify and assess anomalous data points. Surprisingly, no individual data points existed as outliers in this data set. This finding may be attributed to the fact that the data is subject to a team level and not individual performance.

Part III - Methods Applied

Logistic Regression Model

A logistic regression model is used to model the matchup result for the New York Yankees (response variable) with the in-game statistics (predictor variables). The in-game statistics listed in the above table are highly correlated with each other in nature. For example, a score or run in a game is a direct result of a hit. Many statistics in a baseball game can be derived from another. As the goal for the project is drawing inference from the data, applying an elastic-net regularization term bolsters the analysis. The elastic-net will perform variable selection while also dealing with highly correlated predictors at the same time, offering a robust and reliable approach to identify the most relevant predictors influencing the matchup result.

In order to utilize the logistic regression technique and elastic-net to the best degree, the predictors are transformed for this specific portion of the analysis. To perform the analysis, we take the difference in the in-game statistics that are shared between the two teams and use this difference as the predictors in the model. This approach allows the elastic net to select the most relevant variables more clearly, avoiding confusion that could arise if it were to choose variables separately for each team. By using the differences, the model more straightforwardly captures the relative performance, leading to more interpretable and consistent results.

The single baseline logistic regression model analysis will serve multiple purposes. The model accuracy output will help provide credibility for the model and establish if there are any potential data leaks. A good model accuracy translates to having relevant predictors in the model. The inferences we hope to make from the model are only valid if the model itself works. Once accuracy and credibility are established, we can identify the relevant parameters that influence the model the most. While training the model, it was uncovered that multiple predictors in the dataset had to be removed since the logistic regression technique could establish the probability of winning based off those data-leaking predictors. After removing those predictors, the model still performed well but not at an inflated rate (99% accuracy). To further establish credibility for the predictors selected by the elastic-net, bootstrapping techniques are applied to further validate our results.

Bootstrapping

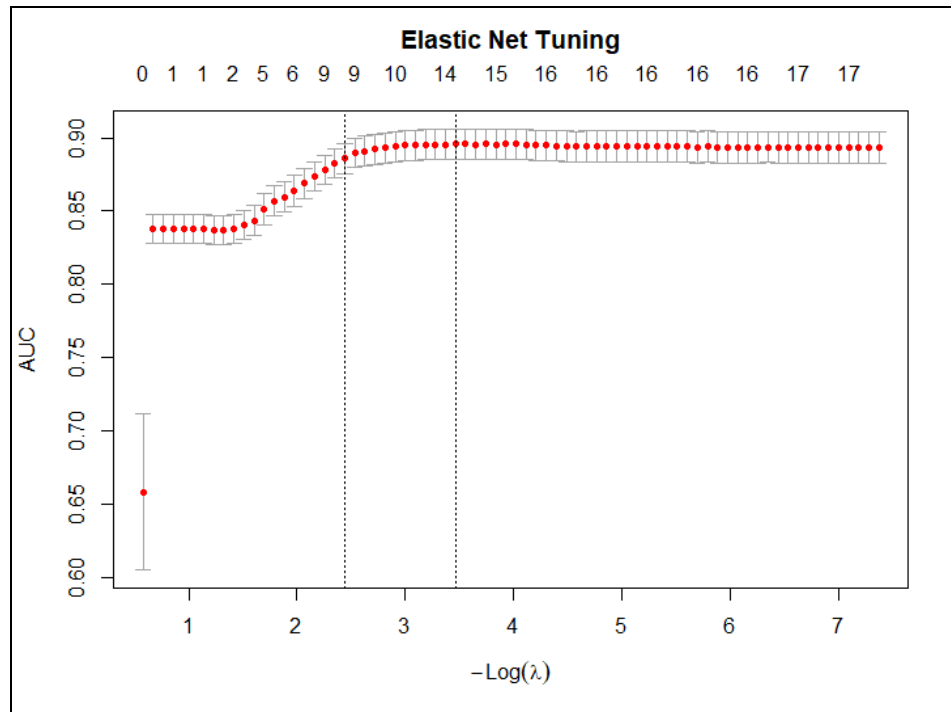
By applying bootstrapping techniques to our logistic regression model, we support then notion that a stable model equates to better inference. We use bootstrapping as a means of validation for the variables selected in the model. If we bootstrap thousands of times and see that the same predictors are repeatedly selected, we have confidence that those predictors are the key contributors to winning a baseball matchup. Crucially, since penalized regression methods like Elastic Net do not provide standard p-values, bootstrapping bridges the gap to inference by generating empirical confidence intervals; this allows us to formally test whether each predictor's impact is statistically distinguishable from zero.

ANOVA & Chi-square Testing

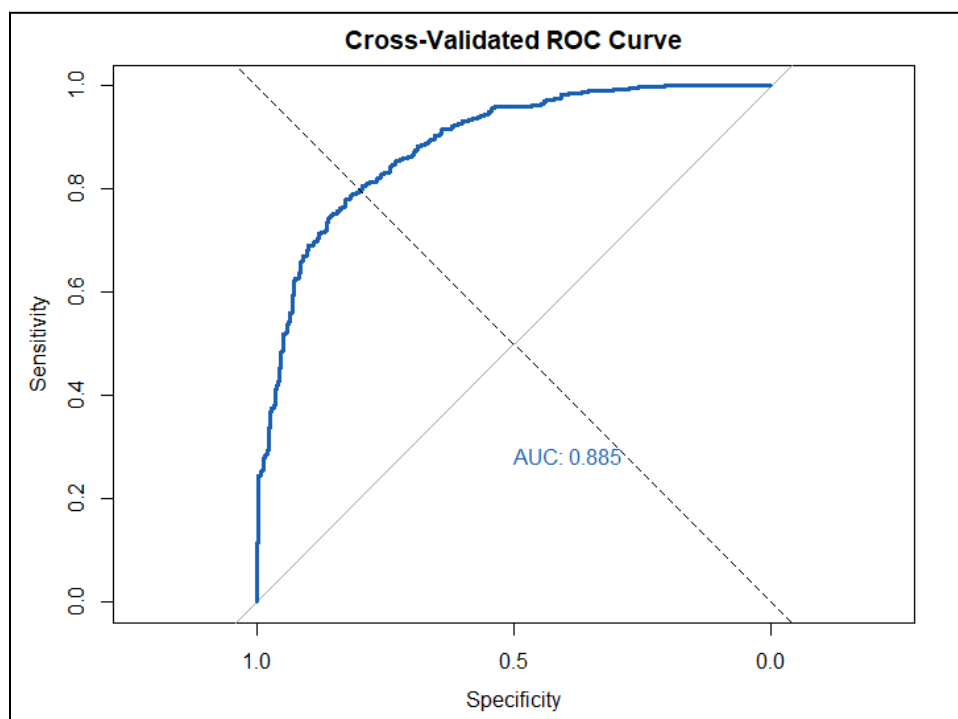
A chi-square test and a series of one-way ANOVA tests were utilized to test if there are any significant differences between games played at home versus away. This analysis addresses whether home field advantage truly exists for the Yankees. The chi-square test was applied to win rate since it is a binary outcome. The ANOVA tests were applied to nine continuous variables: five offensive metrics (team score, team hits, team walks, team strikeouts, team home runs), two defensive metrics (team errors, team double plays), and two opponent metrics (opponent score, opponent strikeouts). A Bonferroni correction ($\alpha = 0.005$) was applied to control for family-wise error rate across the 10 tests. Effect sizes were reported using Cramer's V for the chi-square test and eta-squared for the ANOVA tests to assess practical significance alongside statistical significance. If the results show a significant difference, it suggests that the "home-field advantage" has an impact on team performance.

Part IV - Results

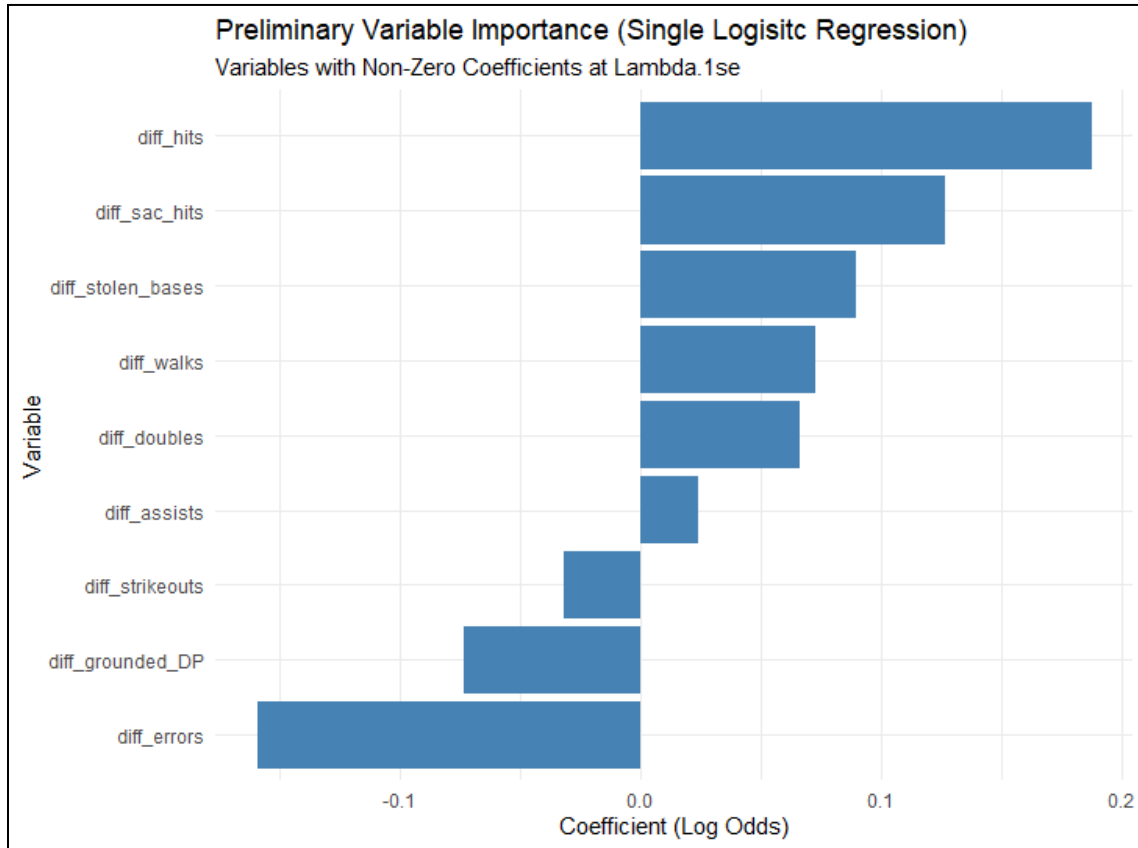
Logistic Regression Results



A K-fold cross validation is applied to the elastic-net to find the optimal tuning parameter. As denoted by the graph, the tuning parameter is able to increase the AUC of the model from about 0.84 to 0.89. The application of K-fold cross-validation validates our variable selection, confirming that the identified statistical drivers are reliable across different subsets of data rather than specific to a single configuration.



The single run of the logistic regression model provides an AUC of 0.885. This confirms that the selected performance metrics are robust explanatory variables that account for the vast majority of the variance in game outcomes.



The Elastic Net model successfully identified a stable subset of nine statistically significant predictors that distinguish winning outcomes from losses.

Offensive Production (Positive Coefficients): Unsurprisingly, the strongest positive associations with winning were found in core offensive metrics.

- **diff_hits**, **diff_doubles**, and **diff_walks** all exhibited positive coefficients, confirming that the primary keys of victory is the ability to generate baserunners and advance them via power. The selection of **diff_walks** alongside hits highlights the importance of plate discipline as a distinct skill from batting kills.

"Small Ball" and Strategy (Positive Coefficients): The model also validated the importance of situational baseball.

- **diff_stolen_bases** and **diff_sac_hits** were selected as positive predictors. This suggests that while raw power drives the majority of wins, the marginal ability to advance

runners into scoring position—either through speed or strategic outs—provides a statistically significant edge, particularly in closer contests.

Defensive Execution (Positive Coefficient)

- **diff_assists** was selected as a positive predictor. This indicates that winning teams generally execute more successful fielding plays (turning ground balls into outs) than their opponents. A positive differential here implies a defense that is active and successfully converting batted balls into outs.

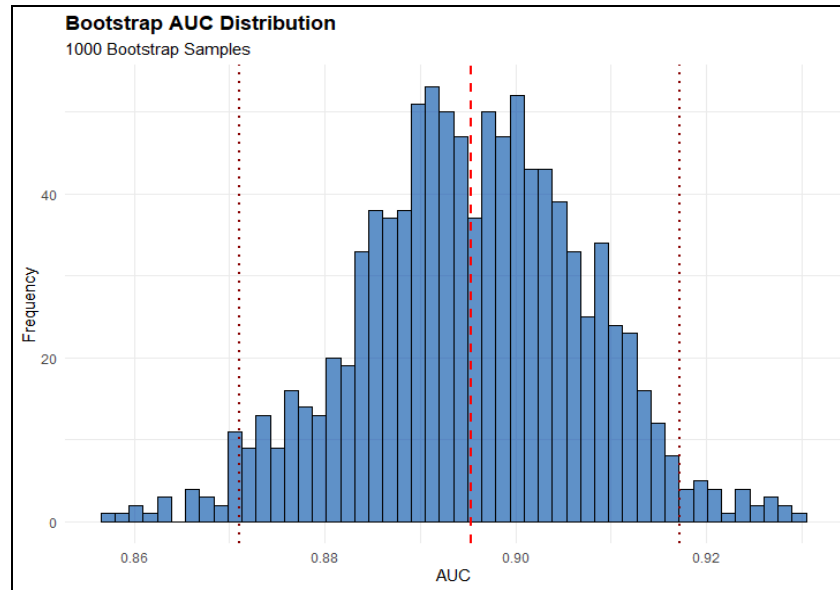
Mistakes (Negative Coefficients) The model identified three specific events that significantly reduce the probability of winning:

- **diff_errors**: The strongest negative driver regarding defense, confirming that giving opponents "free" outs or bases is often fatal.
- **diff_grounded_DP** (Grounded into Double Play): This was penalized heavily by the model, as it erases baserunners and kills offensive momentum twice as fast as a standard out.
- **diff_strikeouts**: A higher differential in batter strikeouts was negatively associated with winning, reflecting the cost of failing to put the ball in play to pressure the defense.

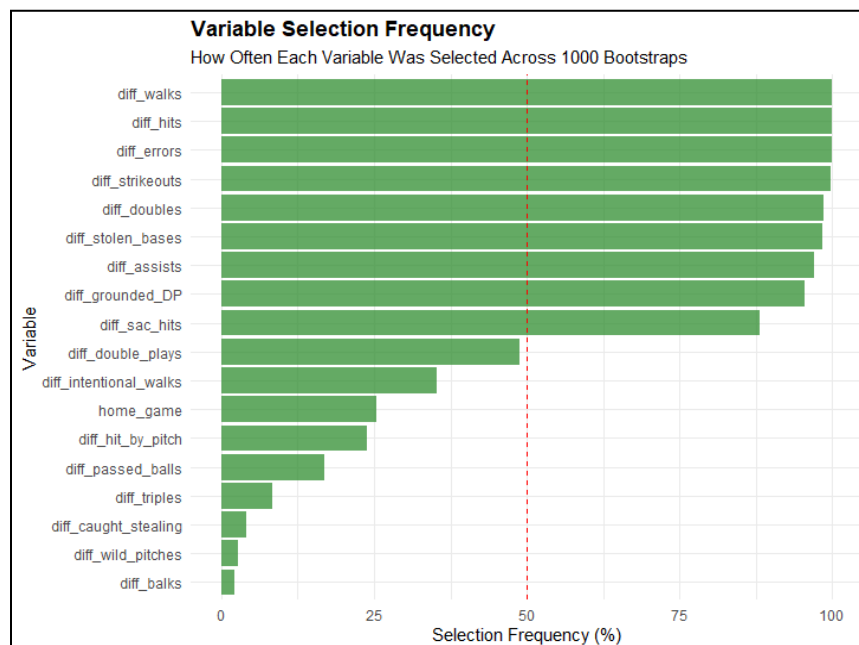
Collectively, the model describes a "winning team" as one that prioritizes traffic on the basepaths (Hits/Walks) and clean execution (Assists/Sac Hits), while actively avoiding momentum-killing mistakes (Double Plays/Errors). The selection of these specific variables over noisier stats (like *Double Plays* or *Balks*) confirms the model's robustness for inference.

Bootstrapping Results

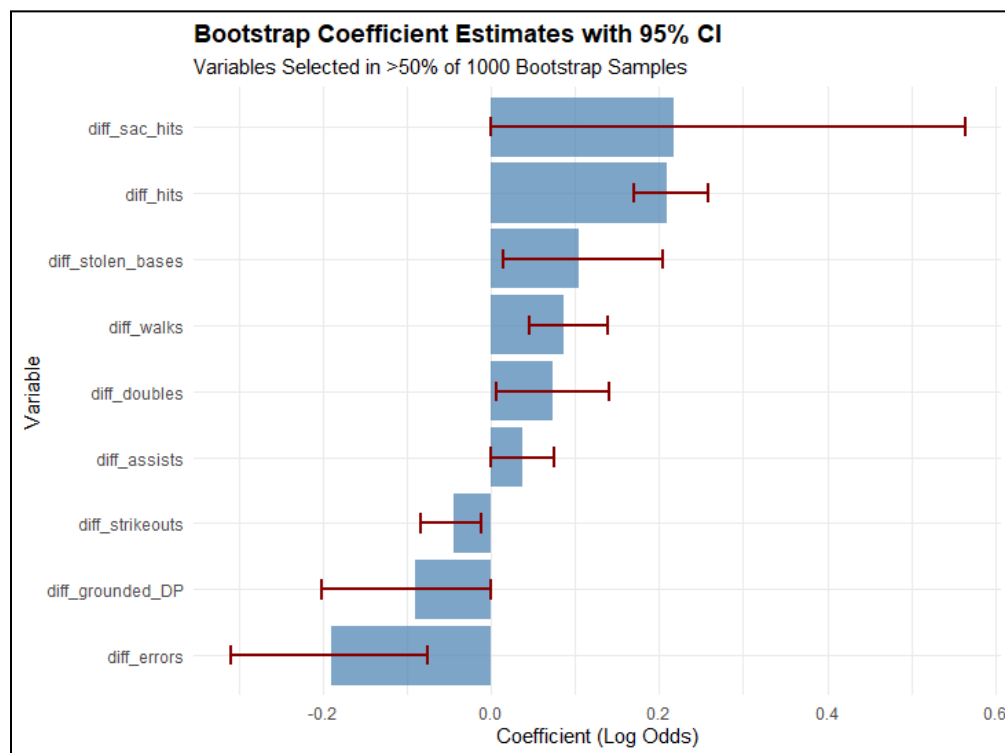
In order to prove that the results from the logistic regression model are meaningful for inference, we apply bootstrapping techniques to the same model. The goal is to showcase that the relationships identified in the baseline model are stable across thousands of resampled datasets, thereby ruling out the possibility that our key variables were selected due to random noise or luck.



The bootstrap analysis shows that the Elastic Net logistic regression model is very good at making predictions. The AUC distribution is tightly centered around 0.89, and the 95% confidence interval is narrow, which means that it works well with different samples. The small range of the bootstrap AUCs indicates that the model works well in general and isn't too sensitive to changes in the sample. Also, predictors that show up consistently over bootstrap resamples with the same sign and little variation show effects that are not affected by sampling noise. This means that they are likely to show real links with the probability of winning rather than random changes. This recurring selection pattern boosts confidence that these variables are structurally significant parts of the game outcome and can be seen as key drivers in the model, not as signals that are unstable or overfit.



The plot of variable selection frequency demonstrates that the model consistently picks a small core group of predictors from the 1,000 bootstrap samples. This means that the inferred drivers of winning are quite stable. In most of the resamples, the variables **diff_walks**, **diff_hits**, **diff_errors**, **diff_strikeouts**, **diff_doubles**, **diff_stolen_bases**, **diff_assists**, **diff_grounded_DP**, and **diff_sac_hits** are chosen. This suggests that they have strong and reliable links to the chance of winning. On the other hand, predictors with selection frequencies well below the 50% criterion are picked inconsistently and probably only give marginal or context-dependent information, which means they are less reliable for making inferences. The clear distinction between frequently and infrequently picked variables bolsters the assertion that the model's principal predictors are not mere artifacts of sampling variability, but rather indicative of consistent, significant effects.



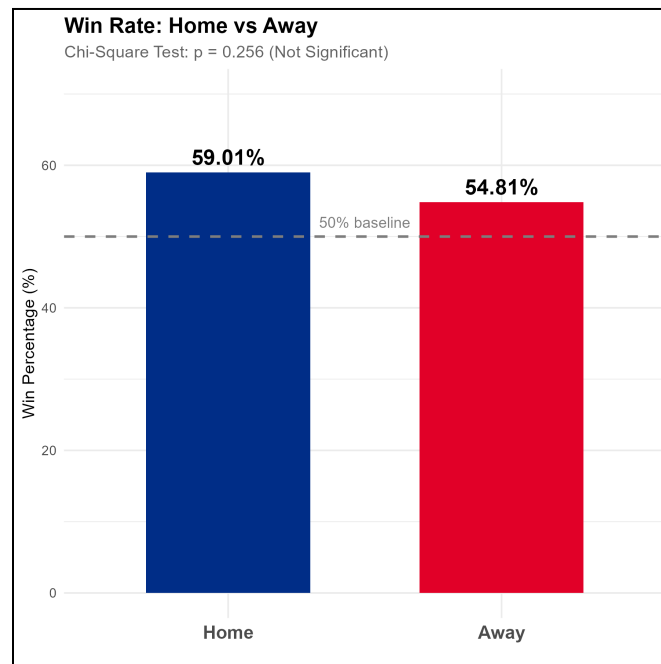
This bootstrap coefficient plot shows predictors that have consistent and stable effects on the log-odds of winning in more than half of the bootstrap samples. The coefficients for measures of offensive advantage like **diff_hits**, **diff_sac_hits**, **diff_stolen_bases**, **diff_walks**, and **diff_doubles**, **diff_assists** are all positive. This means that when these variables go up, the chances of winning also go up. On the other hand, **diff_strikeouts**, **diff_errors** and **diff_grounded_DP** show negative impacts, which means that errors and double plays make it less likely that a team would win. The fact that the coefficient signs are the same and the confidence intervals are quite close throughout resamples suggests that these correlations are strong and not just noise from sampling.

In terms of confidence intervals, the bootstrapped model identified variables such as **diff_hits**, **diff_walks**, **diff_errors**, **diff_strikeouts**, **diff_doubles**, and **diff_stolen_bases** all feature confidence intervals that do not contain zero, confirming with high statistical significance that they have a consistent impact on the outcome. Specifically, because the intervals for hits and walks are entirely positive and 100% non-zero, they represent the most reliable factors of win probability in the model. In contrast, **diff_sac_hits**, **diff_grounded_DP**, and **diff_assists** have intervals that touch or include zero at the boundaries. While these variables show strong average effects, the inclusion of zero suggests that their influence is less stable; in a small percentage of scenarios, these factors may have no discernible impact on the game's result, making them risky predictors that are highly dependent on the specific context of the games sampled.

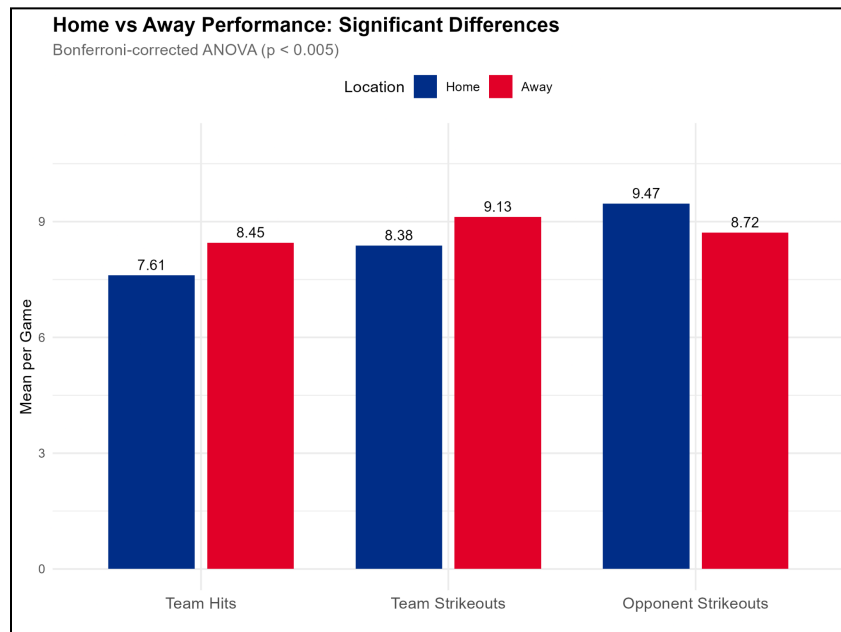
ANOVA & Chi-square Results

A chi-square test and a series of 9 one-way ANOVAs were conducted to analyze whether home field advantage significantly impacts the Yankees' performance metrics and win rate. A Bonferroni correction was applied ($\alpha = 0.005$) to account for multiple comparisons across 10 tests.

The chi-square test analyzed whether win rate differed between home and away games. The ANOVA tests compared home versus away performance across nine key variables: five offensive metrics (team score, team hits, team walks, team strikeouts, team home runs), two defensive metrics (team errors, team double plays), and two opponent metrics (opponent score, opponent strikeouts).



The chi-square test revealed no significant difference in win rate between home (59.0%) and away (54.8%) games ($\chi^2 = 1.29$, $p = 0.256$, Cramer's $V = 0.04$). Despite a 4.2 percentage point difference favoring home games, this gap was not statistically significant per the p-value.



Three performance metrics did display significant home/away differences, these being team hits ($p < 0.001$), team strikeouts ($p < 0.001$), and opponent strikeouts ($p < 0.001$). Notably, Yankees pitchers recorded more strikeouts at home (9.47 vs 8.72 per game), suggesting a pitching advantage. However, the Yankees also recorded fewer hits (7.61 vs 8.45) and fewer strikeouts (8.38 vs 9.13) at home, a counterintuitive pattern that may warrant further investigation.

The remaining six variables showed no significant home/away differences. These include team score ($p = 0.526$), team walks ($p = 0.692$), team home runs ($p = 0.290$), team errors ($p = 0.558$), team double plays ($p = 0.099$), and opponent score ($p = 0.697$). Notably, runs scored were nearly identical at home games versus away games for both the Yankees (4.69 vs 4.83) and their opponents (4.02 vs 4.10). This suggests that while specific in-game behaviors can differ by location, the most crucial determinant for game outcome, scoring runs, remains unaffected by where the game is played.

The sizes of all significant effects were small as indicated by the eta-squared ($\eta^2 < 0.02$), indicating that game location explains less than 2% of variance in any metric. These findings suggest that while home field advantage appears significant in some specific performance metrics, it does not translate to a meaningful increase in overall winning probability for the Yankees.

Part V - Summary

Our analysis identifies a specific "winning formula" driven by nine key metrics that consistently separate victors from losers. The most reliable path to success is built on offensive pressure

and plate discipline; outperforming an opponent in hits and walks are the most certain predictors of a win, supplemented by the consistent advantages of doubles and stolen bases. Conversely, the model shows that winning is equally dependent on avoiding self-inflicted mistakes, with **errors** and **strikeouts** acting as the most reliable negative drivers. While these core factors are universally important, situational plays—like **sacrifice hits**, **defensive assists**, and **double plays**—also play a significant role, though their impact can vary depending on the specific flow of the game. To ensure these findings weren't just a result of "lucky" data, we stress-tested the model by running it through 1,000 different scenarios; the fact that these same nine variables emerged every time confirms that they are true, reliable drivers of success rather than random flukes.

The analysis of home field advantage revealed that the Yankees win at similar rates whether playing at home (59%) or on the road (55%). While some differences in gameplay were observed, such as pitchers recording more strikeouts at home, the actual runs scored by both teams remained nearly identical regardless of home or away status. In short, the home-field advantage does not appear to provide a meaningful edge in terms of winning games for at least the New York Yankees.

Part VI - Future Work

The next major step in improving the model is moving from a single-team study to a league-wide study. The current limitations with this study that the inferences are only valid for the New York Yankees, and shouldn't be generalized for every team in the major league. Future iterations of this study should expand the dataset to include all 30 MLB teams. This would increase the sample size vastly, which would help decrease the overall bias in the study and bring more variability into the interpretation of the results. The greatest undertaking foreseen for this strategy is the massive amounts of data cleaning and preprocessing to be done.

After applying the analysis towards the greater population, the study can be further expanded by using what was learned from the inferential analysis and applying it in a predictive framework. Once there is a strong foundation developed for the key determinants of victory in MLB, we can apply that framework towards forecasting future game outcomes by aggregating the projected performance metrics of starting lineups to generate pre-game win probabilities. The same framework can potentially be translated into a professional setting for MLB personnel such as optimizing roster construction for general managers or developing real-time data driven strategies for coaches in game.

Regarding home field advantage, future work could investigate why the Yankees record more hits away despite also striking out more frequently away. This counterintuitive pattern may reflect differences in opponent quality, field/stadium attributes, or situational hitting approaches. Additionally, analyzing playoff games separately could prove valuable, as home field advantage is commonly believed to be stronger in the playoffs compared to regular season games. Expanding the home field advantage analysis to all 30 MLB teams would reveal whether the patterns observed for the Yankees hold true across the league, or if certain teams benefit more from playing at home than others.